Markov processes and value functions

Answers 1-3: models, transitions and sampled returns

1. Identify the model and the state

(a) In order: **MP, MRP, MDP**.
(b) Room alone is not Markov in this model: the next-location distribution also depends on unrecorded battery information. Include room and battery level in the state, assuming these capture all relevant transition information.

2. Build and use a transition matrix

	Study	Break	Done
Study	0.5	0.25	0.25
Break	0.6	0.4	0
Done	0	0	1

(a) Each row sums to 1.

P(Break,Study) = **0.60**: the probability of Study on the next transition, given Break now.

(b) Sum the three possible intermediate-state contributions:
P(Done after 2 | Study) = (0.50)(0.25) + (0.25)(0) + (0.25)(1)
= 0.125 + 0 + 0.25 = **0.375**. The last term counts first-step arrival at Done followed by its self-loop.

3. Returns, discounting and sample estimates

Episode	Reward sequence	Return, γ = 0.5
1	-2, -2, -2, +10	-2.250
2	-2, -1, -2, -2	-3.250
3	-2, -2, -2, +1, -2	-3.500

(a) Episode 1: -2 - 2(0.5) - 2(0.25) + 10(0.125) = **-2.25**.
Episode 2: -2 - 1(0.5) - 2(0.25) - 2(0.125) = **-3.25**.
Episode 3: -2 - 2(0.5) - 2(0.25) + 1(0.125) - 2(0.0625) = **-3.50**.

(b) Estimated v(C1) = (-2.25 - 3.25 - 3.50) / 3 = **-3.000**. This is a sample estimate, not the exact expectation over all possible trajectories. Use each episode's initial C1 return once.

(c) Episode 1: γ = 0 gives **-2**; γ = 1 gives -2 - 2 - 2 + 10 = **4**. At γ = 0, the later +10 contributes nothing. At γ = 1, it contributes its full value. In both cases, rewards belong to transitions from the listed states, so +10 is earned on Pass → Sleep.

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Answers 4-5: Bellman equations and evaluation updates

4. Evaluate states in the Student MRP

(a) v(C2) = -2 + 0.9[0.8v(C3) + 0.2v(Sleep)]
= -2 + 0.9[0.8(4.087) + 0] = **0.94264 ≈ 0.943**.

(b) v(C3) = -2 + 0.9[0.6v(Pass) + 0.4v(Chat)]
= -2 + 0.9[0.6(10) + 0.4(1.908)] = **4.08688 ≈ 4.087**.

(c) v(FB) = -1 + 0.9[0.9v(FB) + 0.1(-5.013)]
v(FB) = -1 + 0.81v(FB) - 0.45117
0.19v(FB) = -1.45117, so **v(FB) ≈ -7.638**.

(d) C2 has a negative immediate reward, but its discounted expected continuation value is greater than 2. The net expected return is therefore positive. At $\gamma = 0$, **v(C2) = R(C2) = -2**.

Rounding: the lecture shows C2 as 0.9 and C3 as 4.1 to one decimal place. Substituting 4.1 into C2's equation gives 0.952; use the supplied 4.087 for this exercise.

5. Solve a small MRP and practise updates

(a) In order A, B, T:
$R = (-1, 2, 0)^T$; P has rows (0,1,0), (0,0.5,0.5), (0,0,1).
v(A) = -1 + 0.5v(B)
v(B) = 2 + 0.5[0.5v(B) + 0.5v(T)]
v(T) = 0, with terminal value fixed.

(b) Since v(T) = 0, v(B) = 2 + 0.25v(B). Thus
0.75v(B) = 2, so **v(B) = 8/3 ≈ 2.667**.
v(A) = -1 + 0.5(8/3) = **1/3 ≈ 0.333**.

Sweep k	$v_k(A)$	$v_k(B)$	$v_k(T)$
0	0	0	0
1	-1	2	0
2	0	2.5	0
3	0.25	2.625	0

(c) For example, $v_2(A) = -1 + 0.5v_1(B) = 0$. All updates use the previous sweep, not values already updated in the current sweep.

(d) After three sweeps, the estimates are close to, but not equal to, the exact values. These are **fixed-process evaluation** updates. There is no action maximization, so this is not an optimal-control/value-iteration calculation.

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Answer 6: policy evaluation and optimal decisions

6. Evaluate a policy, then improve the decision

(a) G has one action, Finish, with reward +6 followed by terminal T:
$\mathbf{v_\pi(G) = 6}$.
$q_\pi(S,Work) = 0 + 0.5(6) = \mathbf{3}$.
$q_\pi(S,Rest) = 2 + 0.5(0) = \mathbf{2}$.

(b) $v_\pi(S) = 0.25q_\pi(S,Work) + 0.75q_\pi(S,Rest)$
$= 0.25(3) + 0.75(2) = \mathbf{2.25}$.
This averages over the policy's first action. Each action value instead conditions on a particular first action, then follows π.

(c) The policy induces an MRP:
$\mathbf{R^\pi(S) = 0.25(0) + 0.75(2) = 1.5}$.
$\mathbf{P^\pi(G|S) = 0.25}$ and $\mathbf{P^\pi(T|S) = 0.75}$.
Therefore $v_\pi(S) = 1.5 + 0.5[0.25(6) + 0.75(0)] = \mathbf{2.25}$.

(d) At γ = 0.5, choose **Work** at S and Finish at G.
$\mathbf{v_*(S) = max(3,2) = 3}$.
At γ = 0, choose **Rest**, since only its immediate reward +2 matters, while Work gives 0 immediately.

Optional challenge: for general γ, q(S,Work) = 6γ and q(S,Rest) = 2. Work is better when **γ > 1/3**; Rest is better when **γ < 1/3**; they tie at **γ = 1/3**. At the tie, any mixture of the two actions is optimal at S.

Teaching points

A return is associated with a sampled trajectory; a value is an expectation. Immediate rewards can be negative even when long-term values are positive. A policy can be stochastic. Evaluating a fixed policy averages action values using π, whereas finding an optimal value chooses the best achievable action value.

The three small models in Questions 2, 5 and 6 are separate exercises. Do not transfer probabilities or reward conventions between them. Values given for Question 4 are rounded inputs, so minor numerical tolerance is appropriate.

Reference: uploaded RL_part2.pdf, David Silver, Lecture 2: Markov Decision Processes (with supplied annotations), slides 4-30. Practice episodes and the small models were constructed for this worksheet.